

NANYANG TECHNOLOGICAL UNIVERSITY**SEMESTER 2 EXAMINATION 2023-2024****SC3103/CE3103 – EMBEDDED PROGRAMMING**

Apr/May 2024

Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 5 pages.
2. Answer **ALL** questions.
3. This is a closed-book examination.
4. Questions **DO NOT** carry equal marks.
5. **Appendix A** at the end of the exam paper contains schedulability tests that you may find useful while answering some of the questions in this exam.

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1. (a) Consider a real-time embedded system that requires high predictability and responsiveness. Discuss the advantages and disadvantages of implementing this system using a Microkernel architecture compared to a Monolithic Kernel architecture.
(5 marks)
 - (b) Consider a sporadic real-time task set comprising 8 tasks $\tau_1, \dots, \tau_8$, where τ_1 has the highest and τ_8 has the lowest priority. Suppose there are 3 shared resources R_1 , R_2 and R_3 , the usage profile for these resources are as follows ($RAT_{i,j}$ denotes maximum resource access time for resource R_j by task τ_i):
 - $RAT_{2,1} = 4$
 - $RAT_{4,1} = 5, RAT_{4,3} = 3$
 - $RAT_{6,1} = 4$
 - $RAT_{7,2} = 1, RAT_{7,3} = 2$

Note: Question 1 continues on page 2

Any $RAT_{i,j}$ not mentioned above by default has value 0 and it can be assumed that task τ_i does not use resource R_j . Compute the worst-case blocking time of B_i for all the 8 tasks under PCP/SRP.

(10 marks)

- (c) Explain the concept of priority inversion in real-time systems. Provide a detailed example scenario where priority inversion can occur, including the roles of tasks with different priorities. Discuss the potential consequences of priority inversion on system performance and deadlines.

(5 marks)

- (d) Compare independent watchdogs and window watchdogs in embedded systems. Discuss their respective purposes and applications.

(5 marks)

- (e) Describe a situation where a software timer would be a suitable choice for managing timing requirements in a real-time application. Discuss any potential challenges associated with using software timers?

(5 marks)

2. Consider a periodic real-time task system comprising 4 tasks scheduled on a **dual-core** processor. Suppose the tasks in the system are specified as $\tau_1 = (12, 4, 12)$, $\tau_2 = (4, 2, 4)$, $\tau_3 = (3, 1, 3)$ and $\tau_4 = (6, 3, 6)$, where for a task $\tau_i = (T_i, C_i, D_i)$, T_i is the minimum release separation, C_i is the worst-case execution time and D_i is the relative deadline.

- (a) Is this task system feasible on a dual-core processor? Justify your answer.

(5 marks)

- (b) Construct the global EDF schedule for this task system up to time instant 12.

(12 marks)

- (c) Can you conclude from this schedule whether the task system is schedu-

Note: Question 2 continues on page 3

lable under global EDF? Justify your answer.

(3 marks)

3. (a) Consider the following Makefile:

```
all: program

program: main.o utils.o
    gcc $^ -o $@

main.o: main.c
    gcc -c $< -o $@

utils.o: utils.c
    gcc -c $< -o $@
```

Explain the role of each automatic variable used in the Makefile. Discuss the purpose of the 'all' target and how it relates to the 'program' target.

(8 marks)

- (b) What is an orphan process?

(2 marks)

- (c) How does fork() relate to the creation of orphan processes? How does Linux handle these processes?

(5 marks)

- (d) How many 'A' and 'B' are in the output of following code? Explain your answer.

```
int main()
{
    if (fork() + fork() || fork() > 0) {
        printf("A\n");
        fork();
    }
    printf("B\n");
    return 0;
}
```

(10 marks)

4. (a) Briefly describe and compare stream socket and datagram socket.
(6 marks)
- (b) Imagine you execute the command 'ls -l /dev/ttyS0' and receive the following output:
`crw-rw— 1 root dialout 4, 64 Jan 1 00:00 /dev/ttyS0`
Explain the information provided in this output. How are 4 and 64 used to identify and interact with the device?
(6 marks)
- (c) Discuss the usage of lsmod, insmod, rmmod, and modprobe utilities in managing kernel modules.
(8 marks)
- (d) Briefly describe the second stage bootloader in the boot process of embedded linux.
(5 marks)

Appendix A (Schedulability Tests)

In the following equations,

- n denotes total number of tasks in the system,
- m denotes number of cores,
- $\tau_i = (T_i, C_i, T_i)$ denotes a periodic/sporadic task with period T_i , execution time C_i , and relative deadline T_i ,
- B_i denotes maximum blocking time for task τ_i ,
- $HP(\tau_i)$ is the set of all tasks τ_k such that $T_k \leq T_i$, **excluding task τ_i** and
- n_i denotes total number of tasks in $HP(\tau_i)$.

1. **Feasibility test:** $\sum_{i=1}^n C_i/T_i \leq m$

2. **Sufficient test under RM:** $\sum_{i=1}^n C_i/T_i \leq n(2^{1/n} - 1)$

3. **Exact test under RM:** For each task τ_i , $R_i \leq T_i$, where R_i is the response time of task τ_i given by the solution of the following recursive equation

- $R_i(0) = C_i$, and
- $R_i(x) = C_i + \sum_{\tau_k: \tau_k \in HP(\tau_i)} \left\lceil \frac{R_i(x-1)}{T_k} \right\rceil C_k$

4. **Exact test with blocking under EDF:** For each task τ_i , $\sum_{\tau_k: \tau_k \in HP(\tau_i)} C_k/T_k + C_i/T_i + B_i/T_i \leq 1$

5. **Sufficient test with blocking under RM:** For each task τ_i , $\sum_{\tau_k: \tau_k \in HP(\tau_i)} C_k/T_k + C_i/T_i + B_i/T_i \leq (n_i + 1)(2^{1/(n_i+1)} - 1)$

6. **Exact test with blocking under RM:** For each task τ_i , $R_i \leq T_i$, where R_i is given by the solution of the following recursive equation

- $R_i(0) = C_i + B_i$, and
- $R_i(x) = C_i + B_i + \sum_{\tau_k: \tau_k \in HP(\tau_i)} \left\lceil \frac{R_i(x-1)}{T_k} \right\rceil C_k$

END OF PAPER

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Please read the following instructions carefully:

- 1. Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.